CSE111: Programming Language II (Lab)
Spring 2026
Lab Exam - 02
Duration: 25 Minutes

A

Name:	ID:	Section:

Question 1 [C02] [10 Points]

Design a **BurgerHub** class based on the provided tester code and expected output. A **central stock of 500 burgers** is maintained and can be distributed among a **maximum of three branches**. **Each burger** costs **Tk. 150**.

Tester Code

```
public class BurgerHubTester {
  public static void main(String[] args) {

      BurgerHub.viewAll()

      System.out.println("==== 1 ====");
      System.out.println("Are 100 burgers
      available? " + BurgerHub.check(100));

      System.out.println("==== 2 ====");
      BurgerHub b1 = new BurgerHub("Dhanmondi");
      BurgerHub b2 = new BurgerHub("Gulshan");
      b1.viewSummary();

      System.out.println("==== 3 ====");
      b1.sell(5);
      BurgerHub.add(b1, 400);
      b1.sell(20);
      BurgerHub.add(b2, 150);
      BurgerHub.add(b2,40);
      b2.sell(15);

      System.out.println("==== 4  ====");
      BurgerHub.viewAll();

  }
}
```

Expected Output

```
Total Branches: 0
Central Stock: 500
Total Sell: Tk. 0.0
==== 1 ====
Are 100 burgers available? true
==== 2 ====
Branch ID: 1
Branch Name: Dhanmondi
Branch Stock: 0
Branch Sell: Tk. 0.0
==== 3 ====
Low in stock
Burgers added
20 Burgers sold
Cannot add beyond capacity
Burgers added
15 Burgers sold
==== 4 ====
Total Branches: 2
Central Stock: 60
Total Sell: Tk. 5250.0
==== Dhanmondi ====
Branch ID: 1
Branch Name: Dhanmondi
Branch Stock: 380
Branch Sell: Tk. 3000.0
==== Gulshan ====
Branch ID: 2
Branch Name: Gulshan
Branch Stock: 25
Branch Sell: Tk. 2250.0
```

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B

Name:	ID:	Section:

Question 1 [C02] [10 Points]

Design a **BookHive** class based on the provided test code and expected output. The system maintains a **central stock of 300 books** that can be allocated to a **maximum of three outlets**. **Each book** is **issued** at **Tk. 35**.

Tester Code

```
public class BookHiveTester {
  public static void main(String[] args) {

    BookHive.showAll()

    System.out.println("==== 1 ====");
    System.out.println("Are 301 books available?
"+ BookHive.inStock(301));

    System.out.println("==== 2 ====");
    BookHive o1 = new BookHive("Badda");
    BookHive o2 = new BookHive("Mirpur");
    o1.showDetails();

    System.out.println("==== 3 ====");
    o1.issue(40);
    BookHive.allocate(410, o1);
    BookHive.allocate(100, o1);
    o1.issue(70);
    BookHive.allocate(30, o2);
    o2.issue(20);

    System.out.println("==== 4  ====");
    BookHive.showAll();

  }
}
```

Expected Output

```
Total Outlets: 0
Available books in System: 300
Total Earnings: Tk. 0.0
==== 1 ====
Are 301 books available? false
==== 2 ====
Outlet ID: 1
Outlet Name: Badda
Books Assigned: 0
Outlet Earnings: Tk. 0.0
==== 3 ====
Insufficient books in stock
Cannot exceed allocation limit
Books allocated
70 Books issued
Books allocated
20 Books issued
==== 4  ====
Total Outlets: 2
Available books in System: 170
Total Earnings: Tk. 3150.0
==== Badda ====
Outlet ID: 1
Outlet Name: Badda
Books Assigned: 30
Outlet Earnings: Tk. 2450.0
==== Mirpur ====
Outlet ID: 2
Outlet Name: Mirpur
Books Assigned: 10
Outlet Earnings: Tk. 700.0
```

Tentative Solution for Set-A

```java
public class BurgerHub {

  public static int total_branches = 0;
  public static int available_burgers = 500;
  public static double total_all_sell = 0;

  public static BurgerHub branch_list[] = new BurgerHub[3];

  public String branch_name;
  public int branch_id;
  public int total_burgers_available;
  public double total_sell;

  public BurgerHub(String name) {
    branch_name = name;
    branch_list[total_branches++] = this;
    branch_id = total_branches;
  }

  public void sell(int amount) {
    if (total_burgers_available - amount >= 0) {
      total_burgers_available -= amount;
      total_sell += amount * 150;
      total_all_sell += amount * 150;
      System.out.println(amount+" Burgers sold");
    } else {
      System.out.println("Low in stock");
    }
  }

  public static boolean check(int amount) {
    return available_burgers >= amount ;
  }

  public static void add(BurgerHub b, int quantity) {
    if (check(quantity)) {
      b.total_burgers_available += quantity;
      available_burgers -= quantity;
      System.out.println("Burgers added");
    } else {
      System.out.println("Cannot add beyond capacity");
    }
  }

  public void viewSummary() {
```

```
    System.out.println("Branch ID: " + branch_id);
    System.out.println("Branch Name: " + branch_name);
    System.out.println("Branch Stock: " + total_burgers_available);
    System.out.println("Branch Sell: Tk. " + total_sell);
  }

  public static void viewAll() {
    System.out.println("Total Branches: "+total_branches);
    System.out.println("Cenrtal Stock: " + available_burgers);
    System.out.println("Total Sell: Tk. " + total_all_sell);
    for (int i = 0; i < total_branches; i++) {
      System.out.println("==== " + branch_list[i].branch_name + "
====");
      branch_list[i].viewSummary();
    }
  }
}
```

Rubric for Set A

SL	Points to Meet	Marks
1	Proper declaration and implementation of static and instance variables	1.5+0.5
2	Correct implementation of the constructor	1
3	Accurate condition checking and updates to all variables in the sell() method	1.5
4	Correct implementation of the static methods, check() and add(), with proper logic handling	3.5
5	Correct implementation of the output methods, viewSummary() and viewAll(), with proper logic handling	2
Total Marks = 10 Marks		

Tentative Solution for Set-B

```java
public class BookHive {

  public static int total_outlets = 0;
  public static int available_books = 300;
  public static double total_earnings = 0;

  public static BookHive outlet_list[] = new BookHive[3];

  public String outlet_name;
  public int outlet_id;
  public int books_in_outlet;
  public double outlet_earnings;

  public BookHive(String name) {
    outlet_name = name;
    outlet_list[total_outlets++] = this;
    outlet_id = total_outlets;
  }

  public void issue(int amount) {
    if (books_in_outlet - amount >= 0) {
      books_in_outlet -= amount;
      outlet_earnings += amount * 35;
      total_earnings += amount * 35;
      System.out.println(amount + " Books issued");
    } else {
      System.out.println("Insufficient books in stock");
    }
  }

  public static boolean inStock(int amount) {
    return available_books >= amount;
  }

  public static void allocate(int quantity, BookHive b) {
    if (inStock(quantity)) {
      b.books_in_outlet += quantity;
      available_books -= quantity;
      System.out.println("Books allocated");
    } else {
      System.out.println("Cannot exceed allocation limit");
    }
  }
```

```
  public void showDetails() {
    System.out.println("Outlet ID: " + outlet_id);
    System.out.println("Outlet Name: " + outlet_name);
    System.out.println("Books Assigned: " + books_in_outlet);
    System.out.println("Outlet Earnings: Tk. " + outlet_earnings);
  }

  public static void showAll() {
    System.out.println("Total Outlets: " + total_outlets);
    System.out.println("Available books in System: " + available_books);
    System.out.println("Total Earnings: Tk. " + total_earnings);

    for (int i = 0; i < total_outlets; i++) {
      System.out.println("==== " + outlet_list[i].outlet_name + "
====");
      outlet_list[i].showDetails();
    }
  }
}
```

Rubric for Set B

SL	Points to Meet	Marks
1	Proper declaration and implementation of static and instance variables	1.5+0.5
2	Correct implementation of the constructor	1
3	Accurate condition checking and updates to all variables in the issue() method	1.5
4	Correct implementation of the static methods, inStock() and allocate(), with proper logic handling	3.5
5	Correct implementation of the output methods, showDetails() and showAll(), with proper logic handling	2
Total Marks = 10 Marks		